

Pioneers

It was not until 1799 that a British engineer, Sir George Cayley, understood the principles of flight and applied science to the design of a heavier-than-air flying machine. The early pioneers journeyed down many dead ends, designing aircraft with the emphasis more on stability than control, before the Wright brothers finally conquered the air in 1903.



◁ **Henson & Stringfellow Aerial Carriage model 1843**

Origin UK
Engine Steam engine
Top speed N/A

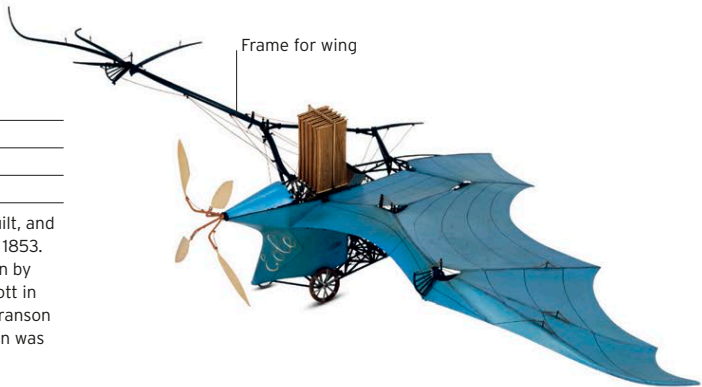
Patented in 1842 the Aerial (as it was known) was designed to be a monoplane with an impressive 148-ft (45-m) wingspan. It was doomed by its poor power-to-weight ratio.



◁ **Cayley Glider 1849**

Origin UK
Engine None
Top speed N/A

Sir George Cayley designed, built, and flew his man-powered glider in 1853. A replica was successfully flown by famous glider pilot Derek Piggott in 1973, and also by Sir Richard Branson in 2003, proving that the design was essentially airworthy.

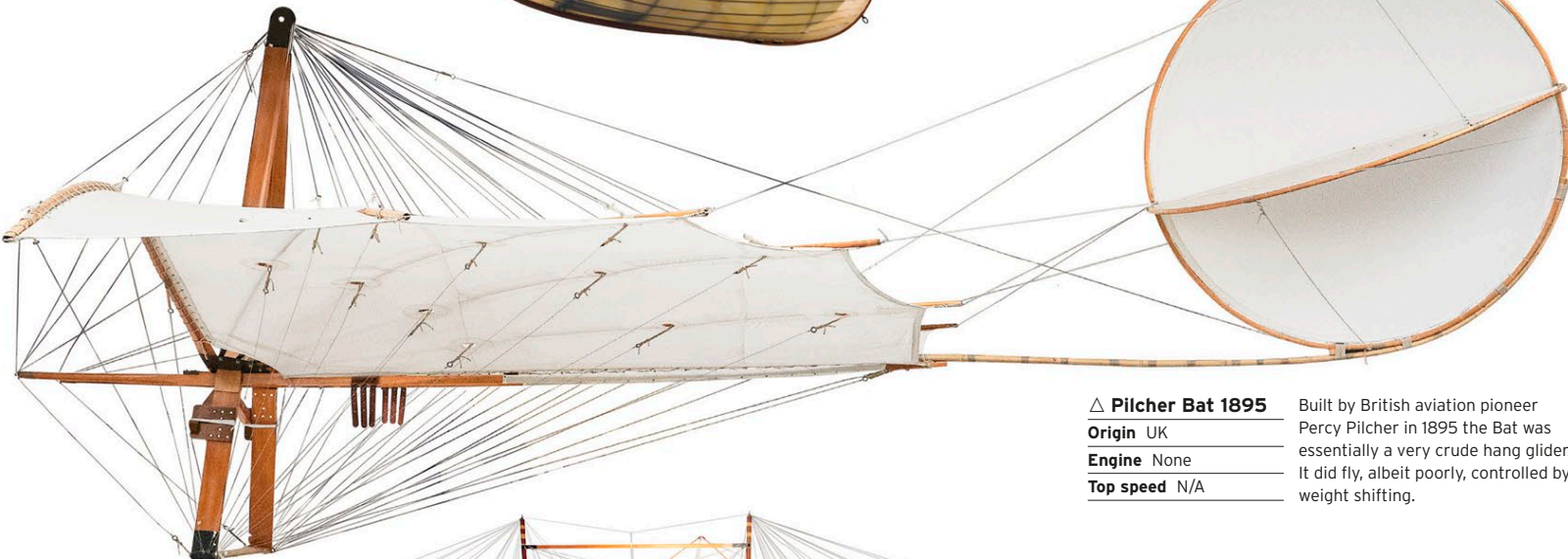


△ **Ader Éole 1890**

Origin France
Engine 20 hp Ader alcohol-burning steam engine
Top speed N/A

This early aircraft was powered by a steam engine. It allegedly achieved a short hop in 1890, but it cannot be considered a successful airplane. The pilot had no directional controls and the heavy power-to-weight ratio ensured that it was a technological dead end.

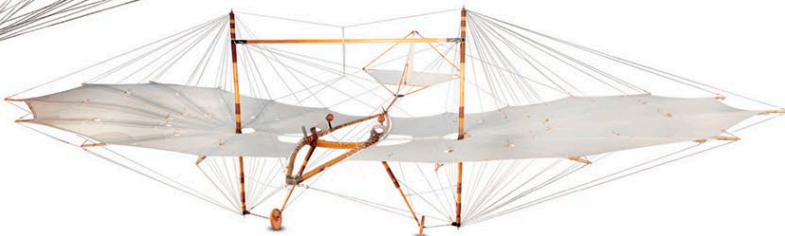
▽ **Biot-Massia Planeur 1879** Early pioneers often applied either maritime principles to their designs, or looked to ornithology. The Biot-Massia Planeur was an attempt to combine the features of a bird with those of a boat—it did not work.



△ **Pilcher Bat 1895**

Origin UK
Engine None
Top speed N/A

Built by British aviation pioneer Percy Pilcher in 1895 the Bat was essentially a very crude hang glider. It did fly, albeit poorly, controlled by weight shifting.



◁ **Pilcher Hawk 1897**

Origin UK
Engine None
Top speed N/A

The Hawk leaned heavily on input from German pioneer Otto Lilienthal. Pilcher was killed on September 30, 1899, when the tailplane failed while demonstrating the Hawk to investors.